

# Choose your robot's next job

Name: \_\_\_\_\_

Date: \_\_\_\_\_ Period: \_\_\_\_\_

Choose one of five builds, test it and explain how you improved it.

## Choose ONE build

### 1. Parcel delivery robot - full module: 5-8

Carry a paper parcel to a red-line delivery space.

### 2. Drawing robot - full module: 9-12

Draw a short line on a protected paper surface.

### 3. Paper sweeper - full module: 13-16

Push three light paper pieces into a collection area.

### 4. Ball-pushing robot - full module: 17-20

Guide one light paper ball toward a goal.

### 5. Rescue carrier - full module: 21-24

Tow a light paper figure on a small rescue sled.

Your teacher will give you the pages for one build. Class 1: choose and build. Class 2: test, improve and share. The page numbers above refer to the complete module workbook.

# Plan before you build

Our chosen job:

Why we chose it:

## LOOK AT THE IDEA



Keep your notes. Change one thing. Try again.

## Words to know

Attachment: A part you add to the robot for a new job.

Evidence: Measurements and notes that show what happened.

Handover: Directions that help someone else use your robot.

**Draw the attachment and label where it joins the base.**

- ☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

## Choice 3: Paper sweeper

Name: \_\_\_\_\_

Date: \_\_\_\_\_ Period: \_\_\_\_\_

Push three light paper pieces into a collection area.

### LOOK AT THE IDEA



Fold the top edge back.

Cut soft bristles below the fold.

Keep the wheels clear.

### Gather your materials

- The shared checked base
- A thin card strip about 12 by 5 cm
- Three tissue-paper squares, each about 2 cm wide
- Removable tape, ruler and scissors

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

## Build your attachment

- 1 Cut a card strip as wide as the robot, but no wider than the clear test lane. Fold its top 2 cm backward to make a mounting edge.
  - 2 Cut short vertical slits about 1 cm apart into the lower 2 cm. These make flexible paper bristles. Round sharp card corners with scissors.
  - 3 Turn the robot off. Attach the folded top edge to approved fixed front mounting points. The flexible ends should just brush the floor without lifting any wheel.
  - 4 Use a clear floor lane. Mark a collection box farther along it. Place three tissue squares in a row across the starting sweep area. Keep them away from wheel paths.
  - 5 Try the short program below. Check that the paper bristles can bend. Stop at once if a piece catches in a wheel; turn the power off before removing it.
  - 6 Count the pieces that reach the box. Repeat three times with the same starting positions. Adjust the bristle height once and compare.
- ☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.